

Pour xgboost :

-> encodage des colonnes : @ip src + @ip dest

-> Scaling : use **scalerXGboost_FeatImportance.joblib**

[had scaler fih 40 feature]

'DST_TO_SRC_SECOND_BYTES', 'MAX_IP_PKT_LEN', 'SRC_TO_DST_SECOND_BYTES' +
dataset w label makach

Ki jou trecuperou les champs m packet :

Voici wach homa les colonnes li yli9 jibouhom :

- DNS_QUERY_TYPE
- TCP_WIN_MAX_OUT
- OUT_PKTS
- MIN_IP_PKT_LEN
- TCP_FLAGS
- PROTOCOL
- TCP_WIN_MAX_IN
- IPV4_SRC_ADDR
- MIN_TTL
- MAX_TTL
- L7_PROTO
- NUM_PKTS_UP_TO_128_BYTES
- L4_DST_PORT
- IPV4_DST_ADDR
- NUM_PKTS_128_TO_256_BYTES
- IN_BYTES
- ICMP_IPV4_TYPE

- RETRANSMITTED_OUT_BYTES
- OUT_BYTES
- DNS_TTL_ANSWER
- DNS_QUERY_ID
- NUM_PKTS_256_TO_512_BYTES
- RETRANSMITTED_IN_PKTS
- NUM_PKTS_512_TO_1024_BYTES
- SRC_TO_DST_AVG_THROUGHPUT
- CLIENT_TCP_FLAGS
- FTP_COMMAND_RET_CODE
- SHORTEST_FLOW_PKT
- RETRANSMITTED_IN_BYTES
- RETRANSMITTED_OUT_PKTS
- IN_PKTS
- LONGEST_FLOW_PKT
- FLOW_DURATION_MILLISECONDS
- DURATION_OUT
- SERVER_TCP_FLAGS
- ICMP_TYPE
- NUM_PKTS_1024_TO_1514_BYTES
- DST_TO_SRC_AVG_THROUGHPUT
- L4_SRC_PORT
- DURATION_IN

RQ:

les champs li ma9dartouch trecupriwhom :

1/utilisation de la valeus la plus fréquente (je pense hiya li tmad
meilleius resulatt)

L4_SRC_PORT	0.0	
L4_DST_PORT	80.0	
PROTOCOL	6.0	
L7_PROTO	0.0	
IN_BYTES	44.0	
IN_PKTS	1.0	
OUT_BYTES	0.0	
OUT_PKTS	0.0	
TCP_FLAGS	0.0	
CLIENT_TCP_FLAGS	0.0	
SERVER_TCP_FLAGS	0.0	
FLOW_DURATION_MILLISECONDS	0.0	
DURATION_IN	0.0	
DURATION_OUT	0.0	
MIN_TTL	64.0	
MAX_TTL	64.0	
LONGEST_FLOW_PKT	1500.0	
SHORTEST_FLOW_PKT	40.0	
MIN_IP_PKT_LEN	40.0	
RETRANSMITTED_IN_BYTES	0.0	
RETRANSMITTED_IN_PKTS	0.0	
RETRANSMITTED_OUT_BYTES	0.0	

RETRANSMITTED_OUT_PKTS	0.0
SRC_TO_DST_AVG_THROUGHPUT	352000.0
DST_TO_SRC_AVG_THROUGHPUT	0.0
NUM_PKTS_UP_TO_128_BYTES	2.0
NUM_PKTS_128_TO_256_BYTES	0.0
NUM_PKTS_256_TO_512_BYTES	0.0
NUM_PKTS_512_TO_1024_BYTES	0.0
NUM_PKTS_1024_TO_1514_BYTES	0.0
TCP_WIN_MAX_IN	0.0
TCP_WIN_MAX_OUT	0.0
ICMP_TYPE	0.0
ICMP_IPV4_TYPE	0.0
DNS_QUERY_ID	0.0
DNS_QUERY_TYPE	0.0
DNS_TTL_ANSWER	0.0
FTP_COMMAND_RET_CODE	0.0

1/**utilisation de lamoyenne** : (testouha malgré machkitch hiya meilleur)

📊 Moyennes des colonnes numériques :

PROTOCOL	1.469054e+01
L7_PROTO	2.878668e+01
IN_BYTES	2.032516e+04
IN_PKTS	1.565667e+01
OUT_BYTES	3.650414e+04
OUT_PKTS	1.392820e+01

TCP_FLAGS	2.781993e+01
CLIENT_TCP_FLAGS	2.459695e+01
SERVER_TCP_FLAGS	1.405810e+01
FLOW_DURATION_MILLISECONDS	9.375297e+05
DURATION_IN	1.461038e+02
DURATION_OUT	2.337782e+01
MIN_TTL	1.046416e+02
MAX_TTL	1.048365e+02
LONGEST_FLOW_PKT	4.472840e+02
SHORTEST_FLOW_PKT	6.601781e+01
MIN_IP_PKT_LEN	3.262552e+01
RETRANSMITTED_IN_BYTES	9.732738e+02
RETRANSMITTED_IN_PKTS	1.206943e+00
RETRANSMITTED_OUT_BYTES	2.609362e+03
RETRANSMITTED_OUT_PKTS	2.275972e+00
SRC_TO_DST_AVG_THROUGHPUT	1.217999e+07
DST_TO_SRC_AVG_THROUGHPUT	1.903501e+07
NUM_PKTS_UP_TO_128_BYTES	2.506621e+01
NUM_PKTS_128_TO_256_BYTES	1.728131e+00
NUM_PKTS_256_TO_512_BYTES	1.513671e+00
NUM_PKTS_512_TO_1024_BYTES	4.190531e-01
NUM_PKTS_1024_TO_1514_BYTES	5.488041e+00
TCP_WIN_MAX_IN	1.133559e+04
TCP_WIN_MAX_OUT	1.186615e+04
ICMP_TYPE	6.613839e+03

ICMP_IPV4_TYPE	2.578831e+01
DNS_QUERY_ID	3.667520e+03
DNS_QUERY_TYPE	1.019490e+01
DNS_TTL_ANSWER	5.978524e+04
FTP_COMMAND_RET_CODE	3.750096e+00

Recuperihom w madouh la1 model :

```
xgboost_modelv3.pkl
```

ymad rsultat numerique

hadi hiya conversion ta3 chaque attack vers string :

```
0 -> Analysis
1 -> Backdoor
2 -> Benign
3 -> Bot
4 -> Brute Force
5 -> DDoS
6 -> DoS
7 -> Exploits
8 -> Fuzzers
9 -> Generic
10 -> Infiltration
11 -> Reconnaissance
12 -> Shellcode
13 -> Theft
14 -> Worms
15 -> injection
16 -> mitm
17 -> password
18 -> ransomware
19 -> scanning
20 -> xss
```

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